Subject: 23CSE301

Lab Session: 09

Notes:

1. Please read the assignment notes carefully and comply to the guidelines provided.
2. Code should be checked into the GitHub. These details shall be provided in the Lab.

Coding Instructions:

1. The code should be modularized; The asked functionality should be available as a function. Please create multiple functions if needed. However, all functions should be present within a single code block, if you are using Jupyter or Colab notebooks.
2. There should be no print statement within the function. All print statements should be in the main program.
3. Please use proper naming of variables.
4. For lists, strings and matrices, you may use your input values as appropriate.
5. Please make inline documentation / comments as needed within the code blocks.
6. **DO NOT USE GenAI tools for coding work for this lab session.**

Main Section:

Please use the data associated with your own project.

Ref:

https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.Perceptron.html

https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.RandomizedSearchCV.html

https://shap.readthedocs.io/en/latest/index.html

https://towardsdatascience.com/explainable-ai-xai-with-shap-regression-problem-b2d63fdca670

https://www.datacamp.com/tutorial/explainable-ai-understanding-and-trusting-machine-learning-models

https://github.com/marcotcr/lime

https://towardsdatascience.com/decrypting-your-machine-learning-model-using-lime-5adc035109b5

https://www.geeksforgeeks.org/introduction-to-explainable-aixai-using-lime/

https://www.datacamp.com/tutorial/explainable-ai-understanding-and-trusting-machine-learning-models

A1. Continue your unfinished experiments from last lab class.

A2. Use cross-validation techniques (***RandomizedSearchCV***()) technique to tune the hyperparameters for your perceptron and MLP networks.

A3. Tabulate your results with various other classifiers such as Support Vector Machines, Decision Tree, RandomForest, CatBoost, AdaBoost, XGBoost, Naïve-Bayes. Tabulate your results for your problem using different performance metrics.

A4. Implement pipeline to allow multiple steps of data processing and classification to be executed simultaneously. Use reference 3 above for pipeline construction and execution.

A5. Using LIME explainer, explain the outcomes of pipeline.

Optional Section:

O1. Study the usage of SHAP (SHapley Additive exPlanations) for usage as explainable tool for ML model behaviour. Use this tool for identifying feature importances for your classification problem.

O2. Study the features of LIME (Local Interpretable Model-agnostic Explanations) tool. Understand the ways the tool can be used for explaining the model behaviour. Use this tool to understand your model's behaviour.

O3. Make performance comparison between RandomSearchCV() & GridSearchCV() for your project.

Report Assignment:

Please update your last week's report in IEEE format. Expand the methodology and results sections with outcomes of this experiments & results obtained. Please discuss your observations, inferences in results & discussion section. Please conclude the report appropriately with these experiments. Consider following points for observation analysis & inferences. Follow the following instructions:

1. Please tabulate the results (don't take screenshots and add to report)
2. Each table and figure should be captioned and cited in text
3. Refer to the "WritingPaper.pptx" file for more instructions and guidelines